NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, Ren Ng

European Conference on Computer Vision (ECCV), 2020

2022. 03. 13

김준철, 고형권, 이예은, 조경진, 김병현

Contents

Background 3D Aware Model

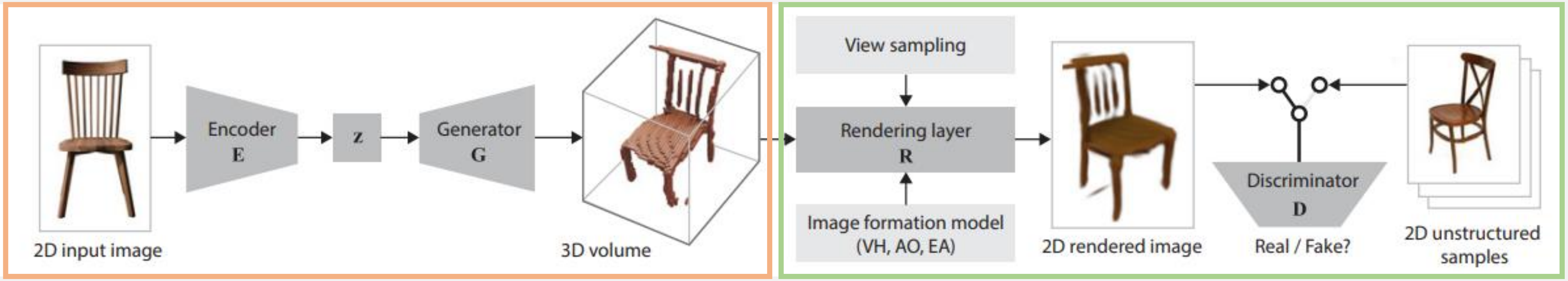
- Several recent works investigate how to control underlying factors of generated images
- Since most of operate in 2D, these fail to disentangle underlying 3D information
- Therefore, most of current methods change the identity of generated images when controlling poses and fail to generate unseen poses which are not in train distribution

Mouth Editing of StyleGAN

Pose Control of GANSpace

Background Computer Graphics

- Digitally synthesizing and manipulating visual content in 3-dimension



➡ Voxel Representation

다양한 방향의 2D 이미지를 합성하여 3D를 표현하는 방법

- Advantage
  - more "accurate" than any other modeling types (e.g. point cloud, mesh-based modeling)
- Disadvantage
  - high computational cost in photorealistic image

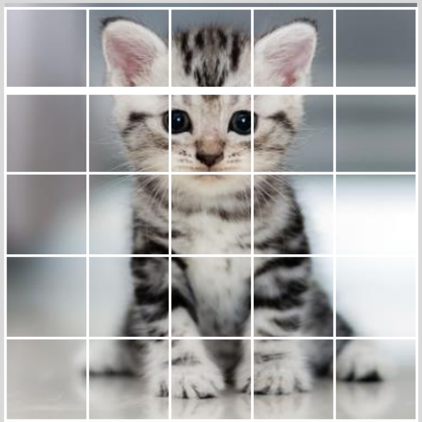
➡ 3D Rendering – Rendering equation

3D 모델로부터 2D 이미지를 생성하는 방법

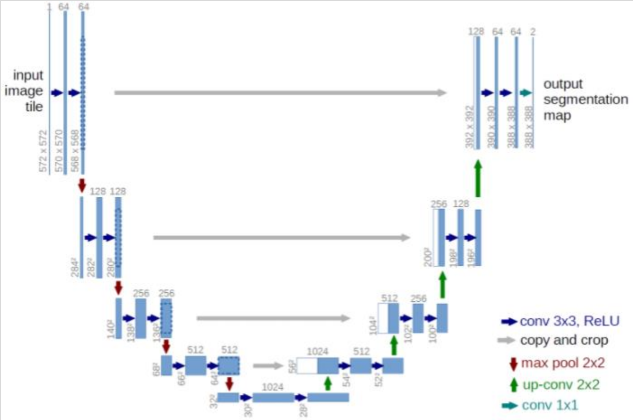
- Volume based rendering
  - MRI, CT
- Surface based rendering

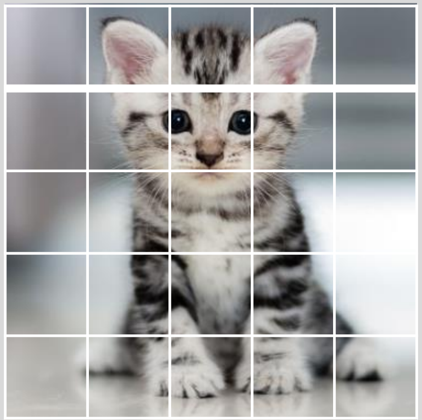
Background Implicit Neural Representation

- Optimizing parameters of a continuous function from discrete data



H x W x C

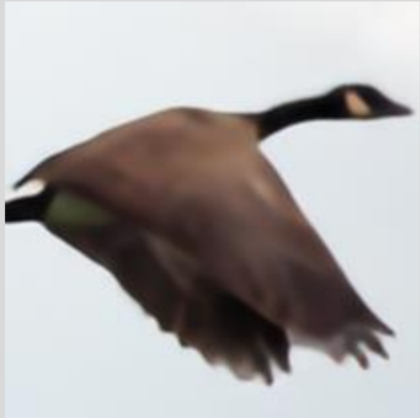




H' x W' x C

x, y

Implicit Neural Representation



R, G, B

F(1, 1) = (255, 0, 0)

F(1, 2) = (0,255,255)

F(1.5, 1.1) = ...?

$\Rightarrow$ $f(x, y) = (r, g, b)$

Benefit of INR

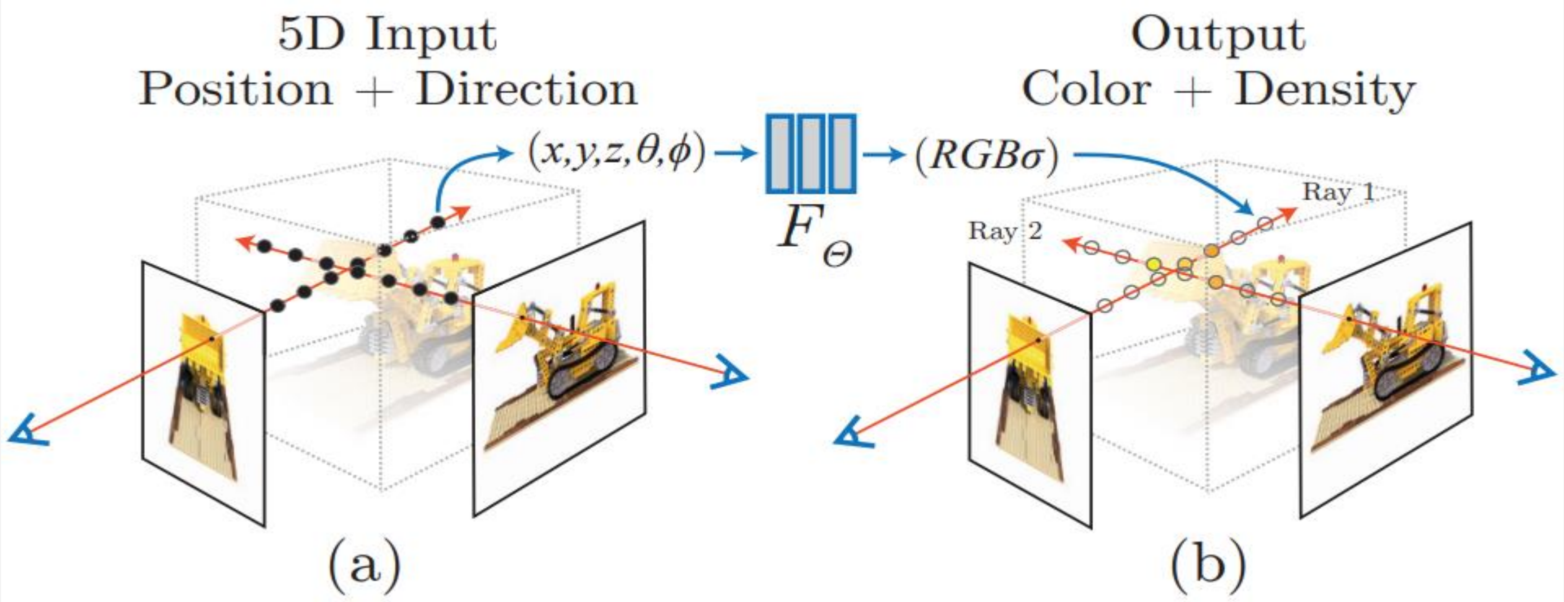
1. Continuous output
2. Light-weight modeling

Introduction

- From 5D spatial point (x, y, z), direction (θ, ϕ) To 4D color (r, g, b), volume density (σ)

$$f(x, y, z, \theta, \phi) = (r, g, b, \sigma)$$

- Method
    1. 레이저를 쏴서 레이저를 지나는 공간상의 3D 포인트로 이루어진 집합을 구성한다.
    2. 3D 포인트 집합과 방향을 모델의 input으로 주어 RGB와 density를 output으로 출력한다.
    3. RGB와 density를 volume rendering 함수에 입력으로 넣고 2D projection을 적용한다.



Introduction

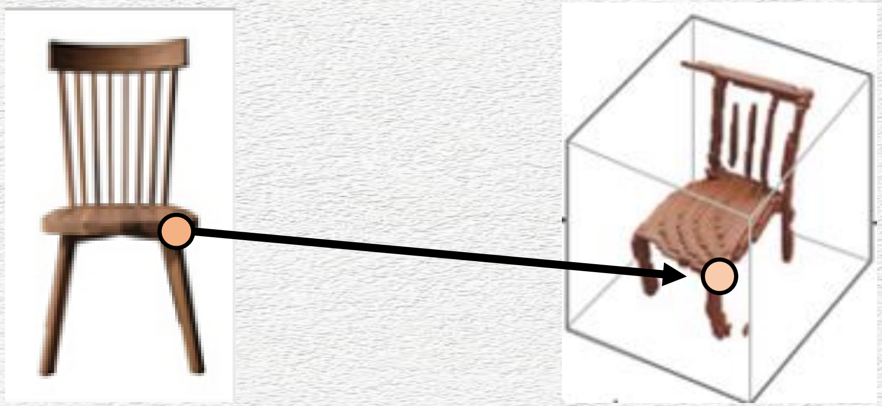
– An approach for representing continuous scenes with complex geometry and materials as 5D neural radiance fields, parameterized as basic MLP networks.

– A differentiable rendering procedure based on classical volume rendering techniques, which we use to optimize these representations from standard RGB images. This includes a hierarchical sampling strategy to allocate the MLP's capacity towards space with visible scene content.

– A positional encoding to map each input 5D coordinate into a higher dimensional space, which enables us to successfully optimize neural radiance fields to **represent high-frequency** scene content.

Related Work

- Computer vision is encoding objects and scenes in the weights of an MLP that directly maps from a 3D spatial location to an implicit representation of the shape



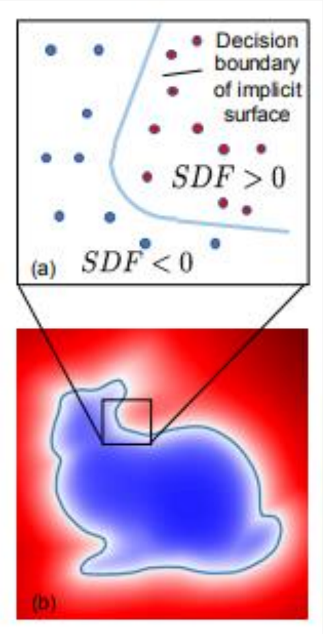
- Limitation : Unable to reproduce realistic scenes with complex geometry with the same fidelity

01 Neural 3D shape representations

02 View synthesis and image-based rendering

Related Work Neural 3D shape representations

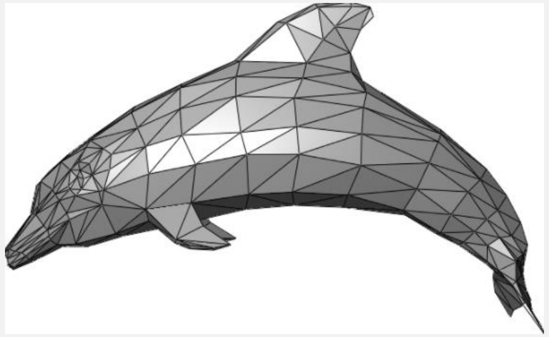
- Previous work
  - Implicit representation of continuous 3D shapes as level sets by optimizing deep networks that map xyz coordinates to signed distance functions.
  - Limit : these models are limited by their requirement of access to ground truth 3D geometry, typically obtained from synthetic **3D shape datasets**.



- Recent work
  - Neural implicit shape representations to be **optimized using only 2D images** with rendering function
    1. 3D occupancy fields and use a numerical method to find the surface intersection for each ray
    2. Differentiable rendering function consisting of a recurrent neural network
  - Limit : Difficult to potentially represent complicated and high resolution geometry

Related Work View synthesis and image-based rendering

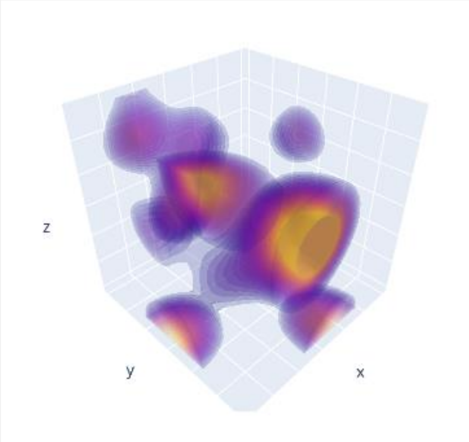
- Surface-based rendering



  - One popular class of approaches uses mesh-based representations of scenes with either diffuse or view-dependent appearance (3D mesh -> 2D image)
  - Limits

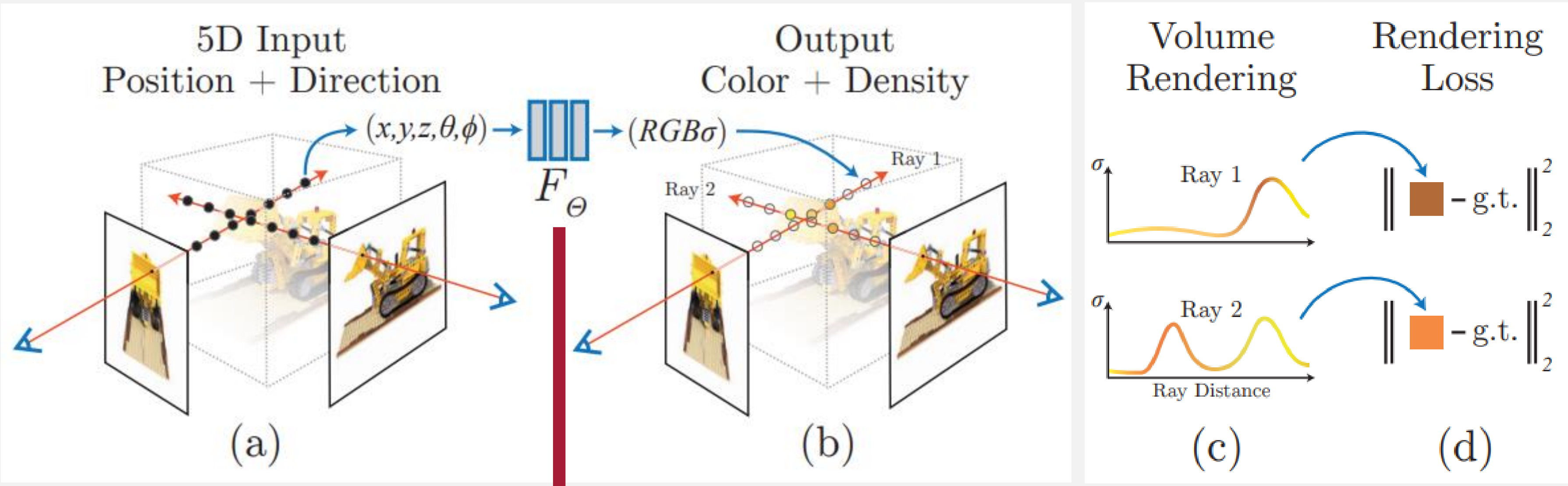
    1. Gradient-based mesh optimization based on image reprojection is often difficult, likely because of local minima or poor conditioning of the loss landscape
    2. Unavailable for unconstrained real-world scenes

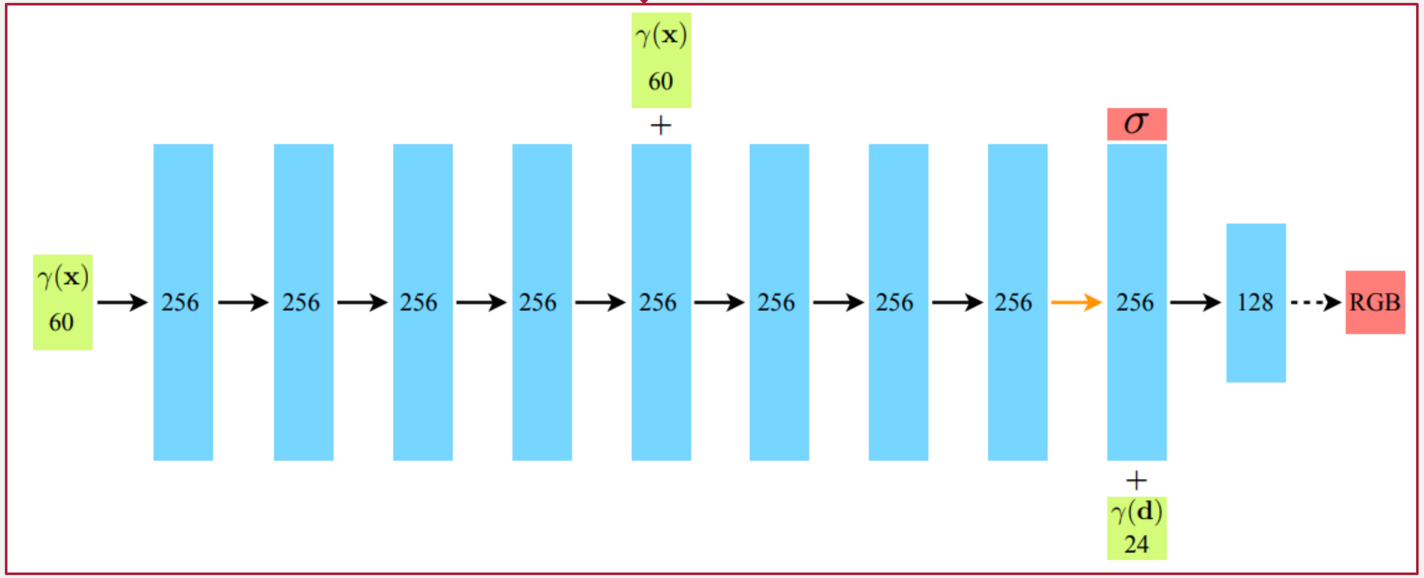
- Volume-based rendering



  - high-quality photorealistic view synthesis from a set of input RGB images
  - Limit : **Poor time and space complexity** due to their discrete sampling

Neural Radiance Field Scene Representation





$$F_\theta : (x, d) \rightarrow (c, \sigma)$$

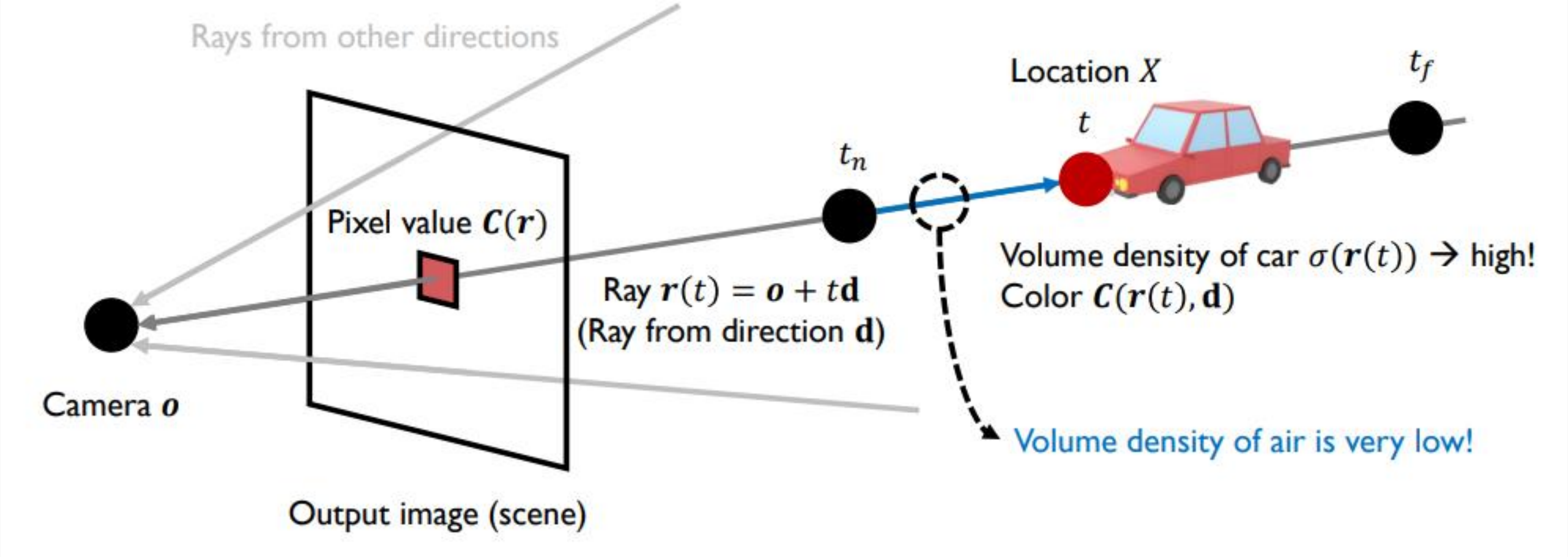
x : 3D location (x, y, z)

d : 2D viewing direction

Volume Rendering with Radiance Fields

The expected color $C(r)$. $r(t) = o + td$ with near and far bound t_n and t_f

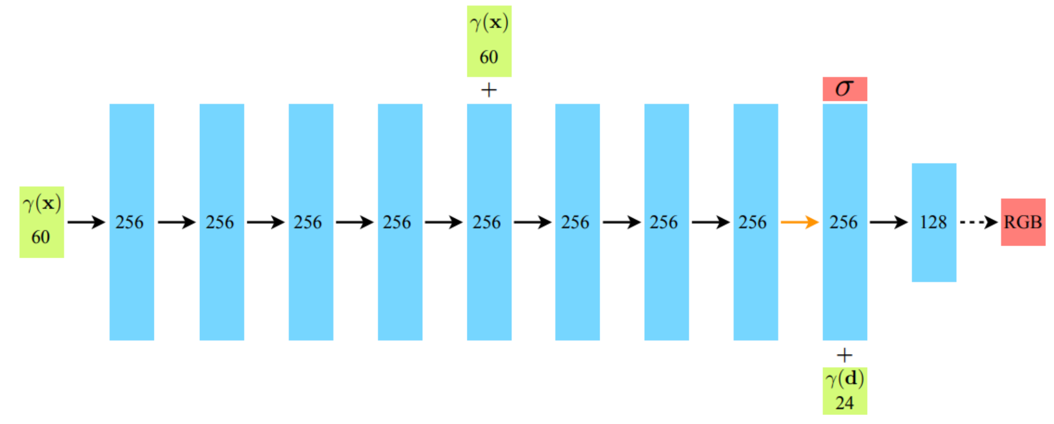
$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})dt\,, \text{ where } T(t) = \exp\left(-\int_{t_n}^{t} \sigma(\mathbf{r}(s))ds\right)$$

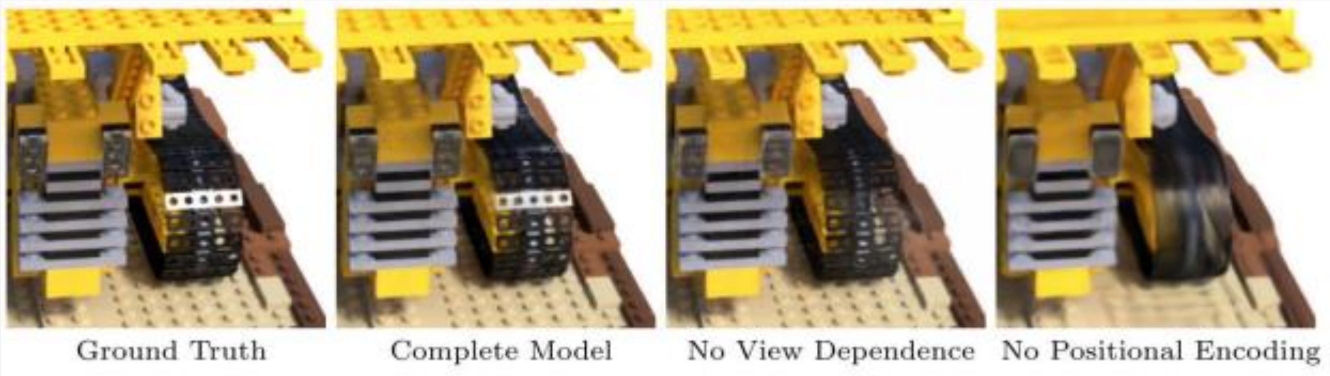


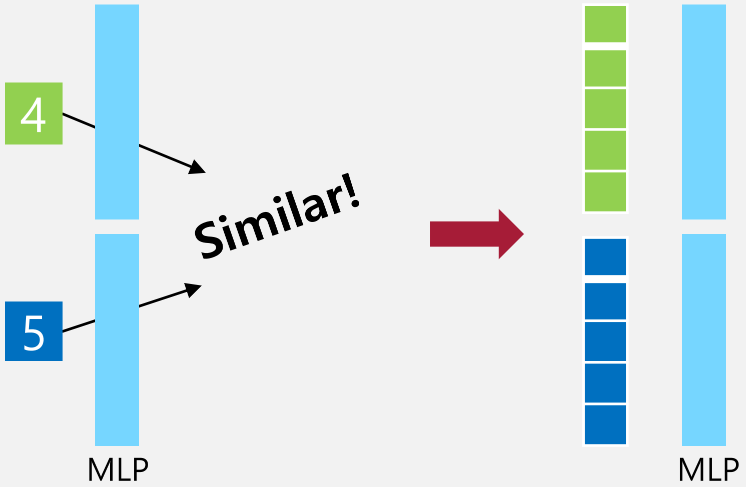
Q & A

Optimizing a Neural Radiance Field Positional encoding

$$\gamma(p) = (\sin(2^0 \pi p), \cos(2^0 \pi p), \dots, \sin(2^{L-1} \pi p), \cos(2^{L-1} \pi p))$$







(x, y, z) 3 dimension -> 60 dimension with L = 20

(θ, ϕ) 2 dimension -> 24 dimension with L = 6

Optimizing a Neural Radiance Field Hierarchical volume sampling

- Computing Integral

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})dt\,, \text{ where } T(t) = \exp\left(-\int_{t_n}^{t} \sigma(\mathbf{r}(s))ds\right)$$

$$\hat{C}_c(\mathbf{r}) = \sum_{i=1}^{N_c} w_i C_i\,, \qquad w_i = T_i(1 - \exp(-\sigma_i \delta_i))$$

$$t_i \sim \mathcal{U}\left[t_n + \frac{i-1}{N}(t_f - t_n),\ t_n + \frac{i}{N}(t_f - t_n)\right]$$

Optimizing Coarse feature - N = 64

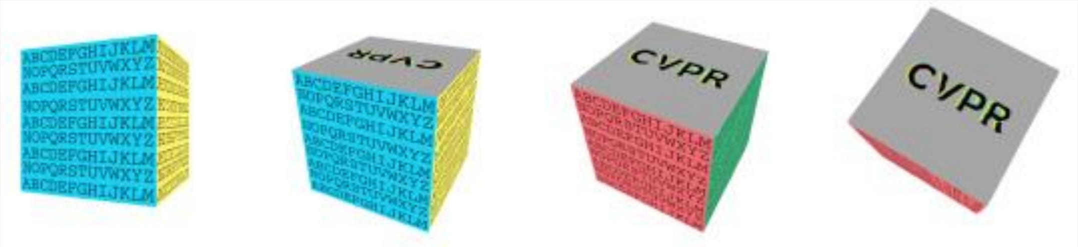
Optimizing Fine feature - N = 128

$$L_{total} = \sum_{r \in R}\left[\left\|\hat{C}_c(\mathbf{r}) - C(\mathbf{r})\right\|_2^2 + \left\|\hat{C}_f(\mathbf{r}) - C(\mathbf{r})\right\|_2^2\right]$$

Results Datasets

- Diffuse Synthetic 360

  512 × 512 pixels , 479 as input and 1000 for testing



- Real Forward-Facing

  1008×756 pixels , 20 to 62 images and 1/8 for the test set

Air Plants

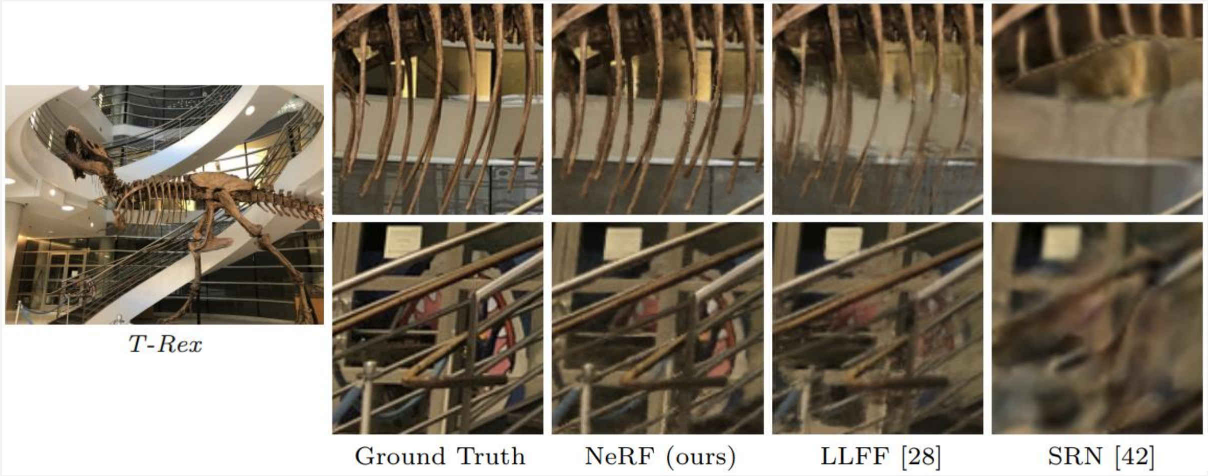
Fern

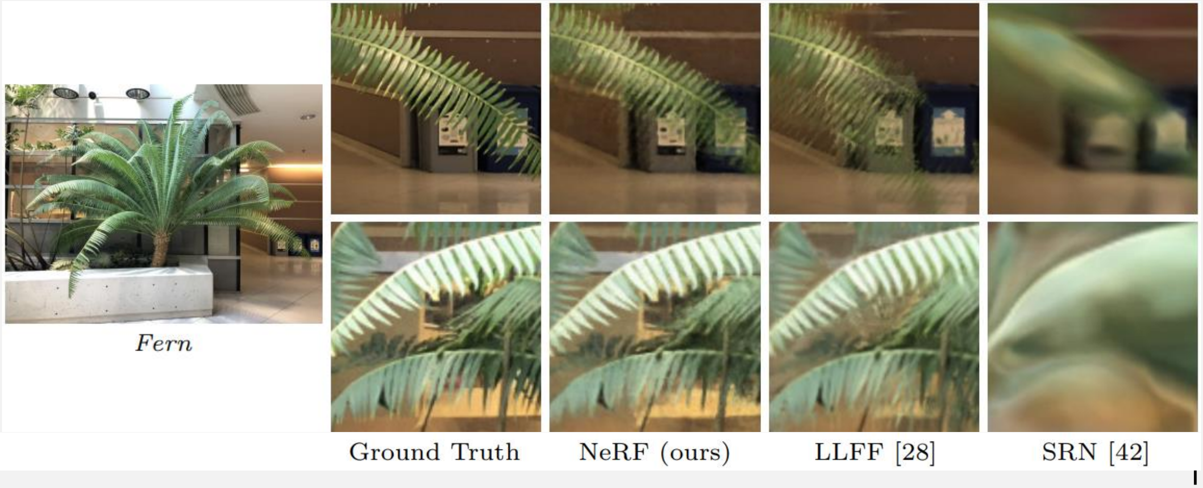
T-Rex

Results Comparisons

- 모델 별 성능 비교

Method	Diffuse Synthetic 360° [41]			Realistic Synthetic 360°			Real Forward-Facing [28]		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
SRN [42]	33.20	0.963	0.073	22.26	0.846	0.170	22.84	0.668	0.378
NV [24]	29.62	0.929	0.099	26.05	0.893	0.160	-	-	-
LLFF [28]	34.38	0.985	0.048	24.88	0.911	0.114	24.13	0.798	**0.212**
Ours	**40.15**	**0.991**	**0.023**	**31.01**	**0.947**	**0.081**	**26.50**	**0.811**	0.250





Results Discussion

Models

- LIFF - Local Light Field Fusion

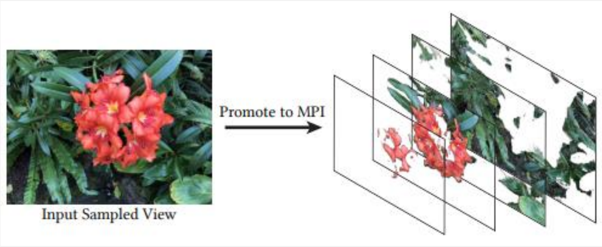
  - Multi plane image(MPI) for each view with 3D convolution > rendering

- SRN - Scene Representation Networks

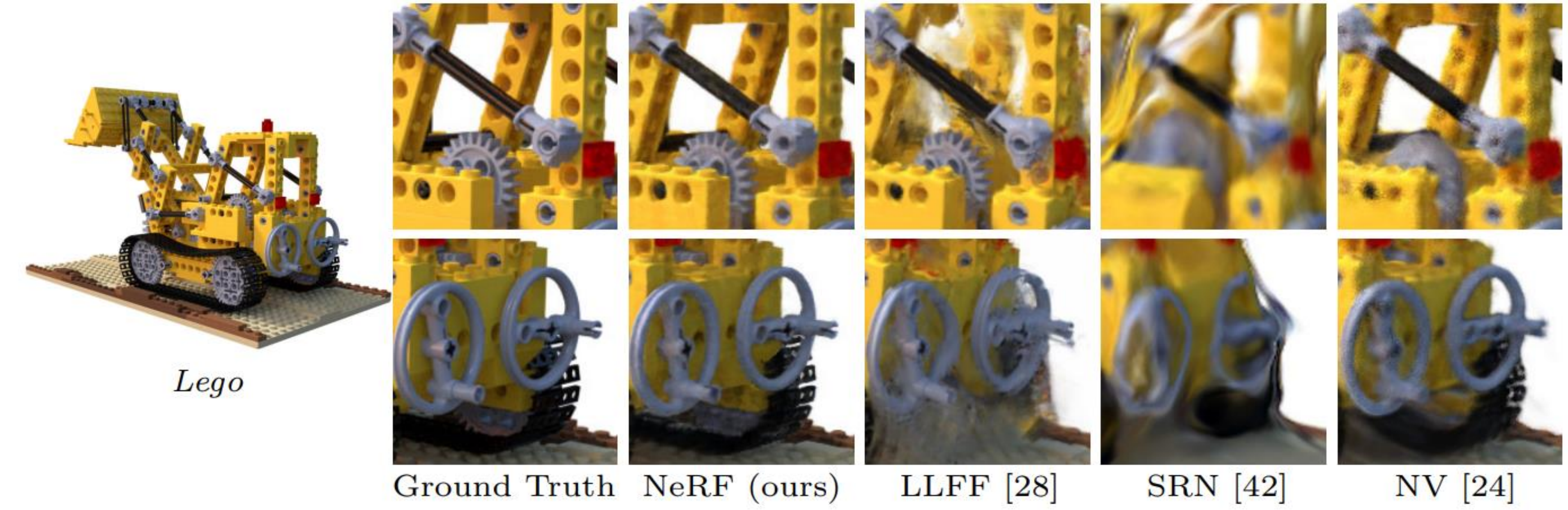
  - INR, F(x, y, z) = Feature vector > color decoding

- NV - Neural Volumes

  - 2D conv to embedding in all images > 3D deconvolution > 128^3 RGBa voxel grid



LIFF



Results Ablation studies

	Input	#Im.	L	(N_c, N_f)	PSNR↑	SSIM↑	LPIPS↓
1) No PE, VD, H	xyz	100	-	(256, -)	26.67	0.906	0.136
2) No Pos. Encoding	$xyz\theta\phi$	100	-	(64, 128)	28.77	0.924	0.108
3) No View Dependence	xyz	100	10	(64, 128)	27.66	0.925	0.117
4) No Hierarchical	$xyz\theta\phi$	100	10	(256, -)	30.06	0.938	0.109
5) Far Fewer Images	$xyz\theta\phi$	25	10	(64, 128)	27.78	0.925	0.107
6) Fewer Images	$xyz\theta\phi$	50	10	(64, 128)	29.79	0.940	0.096
7) Fewer Frequencies	$xyz\theta\phi$	100	5	(64, 128)	30.59	0.944	0.088
8) More Frequencies	$xyz\theta\phi$	100	15	(64, 128)	30.81	0.946	0.096
9) Complete Model	$xyz\theta\phi$	100	10	(64, 128)	**31.01**	**0.947**	**0.081**

PE : positional encoding
VD : view dependence
H : hierarchical sampling

Conclusion

- Produce **better renderings** than the previously-dominant approach of training deep CNN
- Make more rendering more **sample-efficient**
- This work makes progress towards a graphics pipeline based on **real world imagery**
- **5 MB** for the network weights. (LLFF weights over 15GB for one "Realistic Synthetic" scene)

Q & A